PRACTICE SHEET

1. If a, b and c are real numbers, then roots of the equation $(x-a)(x-b)+(x-b)(x-c)+(x-c)(x-a)=0$ are always:
(a) Real (b) Imaginary
(c) Positive (d) Negative

2. If $x = 2 + 2^{1/3} + 2^{2/3}$, then what is the value of $x^3 - 6x^2 + 6x$?
(a) 1 (b) 2
(c) 3 (d) –2

3. For the two equations $x^2 + mx + 1 = 0$ and $x^2 + x + m = 0$, what is/are the value/values of m for which these equations have at least one common root?
(a) Only –2 (b) Only 1
(c) –2 and 1 (d) –2 and –1

4. If (x+a) is a factor of both the quadratic polynomials $x^2 + px + q$ and $x^2 + lx + m$, where p,q, *l* and m are constants, then which one of the following is correct?
(a) $a=\frac{(m-q)}{(l-p)},(l\neq p)$ (b) $a=\frac{(m+q)}{(l+p)},(l\neq p)$
(c) $l=\frac{(m-q)}{(a-p)},(a\neq p)$ (d) $p=\frac{(m-q)}{(a-1)},(a\neq 1)$

5. If the roots of the equation $4\beta^2 + \lambda\beta - 2 = 0$ are of the from $\frac{k}{k+1}$ and $\frac{k+1}{k+2}$ then what is the value of λ?
(a) 2k (b) 7

12. If the equations $x^2 + k^2 = 2(k+1)x$ has equal roots, then what is the value of k?
(a) $-\frac{1}{3}$ (b) $-\frac{1}{2}$
(c) 0 (d) 1

13. If α, β are the roots of the equations $x^2 - 2x - 1 = 0$, then what is the value of $\alpha^2\beta^{-2} + \alpha^{-2}\beta^2$
(a) –2 (b) 0
(c) 30 (d) 34

14. Which one of the following values of x, y satisfies the equation $2x + 3y \le 6$; $x \ge 0$, $y \ge 0$?
(a) x = 0, y = 3 (b) x = 1, y = 2
(c) x = 1, y = 1 (d) x = 4, y = 0

15. What is the value of x at the intersection of $y=\frac{8}{(x^2+4)}=$ and x + y = 2?
(a) 0 (b) 1
(c) 2 (d) –1

16. If α and β are the roots of the equation $x^2 + 6x + 1 = 0$, then what is $|\alpha - \beta|$ equal to?
(a) 6 (b) $3\sqrt{2}$
(c) $4\sqrt{2}$ (d) 12